Business to Workforce (B2W)

"Extending The Enterprise"

Java Users Group

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Kevin Clark
President & CEO
NotWired, Inc.

Business to Workforce (B2W)

"Extending The Enterprise"

- About NotWired
- History and Trends
- Choosing the Right Application
- Choosing the Right Development Strategy
- ROI Equation and Cost Examples
- Myths, Reality and History
- NotWired Products and Services

About NotWired

- Subsidiary of Objectsoft
- Started NotWired development in 1998
- 16 people in engineering and consulting
  - Very strong OO skills
  - Very strong Java skills
  - Very strong embedded and C skills
- NotWired is Delaware C-corp as of 2000

History and Trends

- Historic wireless applications
- Historic wireless characteristics
- Changing Internet market
- Industry trends
- Year 2001 trends

Historic Wireless Applications

- Courier/Messenger
- Utilities/Field Service
- Field Sales/Order Processing
- Law Enforcement
- Dispatch/Logistics
- Healthcare
- Finance/Trading

Historic Wireless Characteristics

- Proprietary solutions
- Expensive
- Large deployments
- Used by mobile workforces
- ROI driven with cost distributed over many workers

Changing Internet Market







Microsoft Internet Explorer

Netscape

Wired

Technology

Java/Servlets
ASP/VB
CGI
HTML/javascript






Microsoft Internet Explorer

JAVA

WAP

phone.com

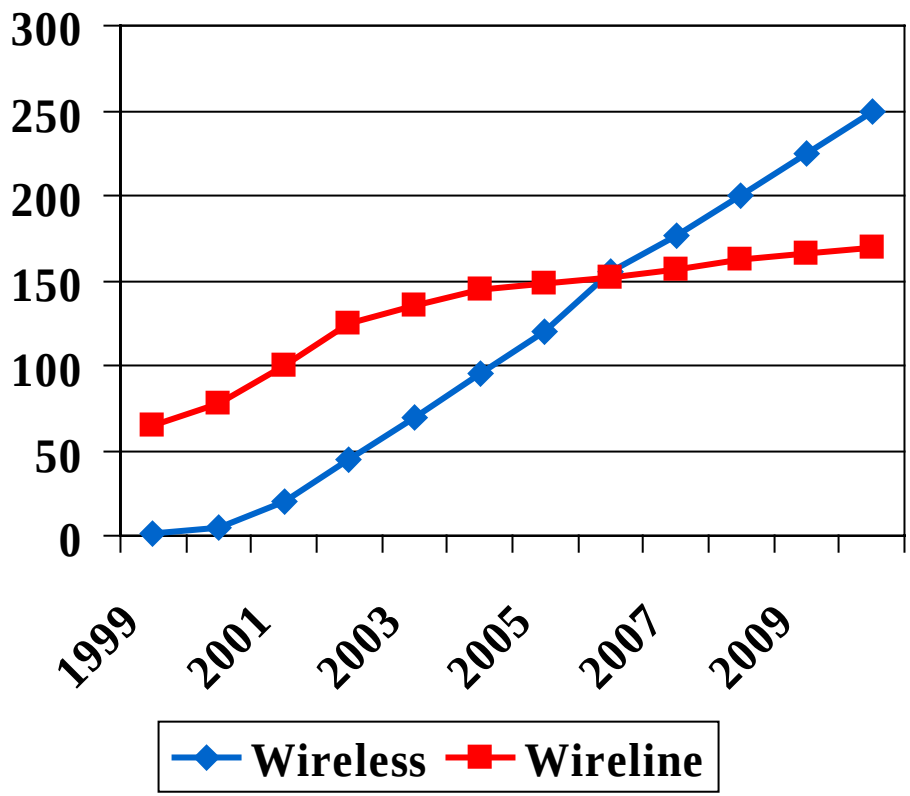
NotWired

New Entrants...

WML, HDML, PQA,
cHTML, J2ME, PalmOS/C,
others...

Industry Trends

- Proliferation of devices including PDAs, smart phones and other devices
- 47 million mobile and remote workers by 2002
- Wireless internet to pass wired internet in 2006



2001 will be "The year for wireless"

- Advancement in wireless technology
  - carriers and technologies
  - standards
  - speed and coverage
  - developer tools and skills
  - pricing plans

2001 will be "The year for wireless"

- Advancement in devices
  - speed, size, battery life and memory
  - color
  - physical interfaces
    - stylus, keyboard, voice, finger, biometrics

2001 will be "The year for wireless"

- Why?
  - Because the cost of devices, access and the availability of skills and tools to support development will make the ROI a positive calculation for many IT development.

Choosing Directions

- Right application
- Right device
- Right strategy
- Right partner

Picking the Right Application

- Profile the section of workforce that needs remote access
  - sales and marketing, customer field service, delivery
- Prioritized information used outside office
- Research device user acceptance
- Understand network and backend issues

Picking the Right Device

- Who will use it?
- What are the applications?
- What is their work environment like?
- How easy is it to use?
- What does it cost?
- Who are the covering carriers?
- Roaming issues?
- Skills to support development?

Picking the Right Device

Laptop	Disconnected or connected design? Keyboard.
WAP phone	Small screen, simple apps.
PalmOS	Local storage and variety of languages. Native apps in C. Stylus.
PocketPC	Local storage and variety of languages. Stylus.
RIM	Local storage. C++, Java and WAP. Keyboard.
Specialized	Some have keyboards, some built for harsh environments.

Picking the Right Strategy Make, Buy or Outsource

- Dynamic marketplace
  - Short development cycles
  - Changing technology
- Underdeveloped Industry
  - Limited skilled resources
  - High costs for R&D
- Device Support
  - Limited software

Picking the Right Strategy Make, Buy or Outsource

- Skill development
  - Is the application core or an add-on?
  - Do you want to deal with all the intricacies of devices, networks, protocols and deployment?
- User acceptance
  - Can you design it well? Will they accept it?
  - Will it roll out to large scale and do you have the skills to support the rollout?

Picking the Right Strategy Make, Buy or Outsource

- Security
  - How much?
  - How will it be integrated into our corporate strategy?

Picking the Right Partner

- Developed on delivery platforms
- Developed more than ‘Hello World’
- Web, OO, database and systems development skills

ROI Equation

- Application development
  - People, tools, testing, hardware, maintenance
- Deployment
  - Hardware and software
  - Updates
- Monthly service fees
  - Hosting, access
- Training

Sample - Remote email

- Give executive team ability to check email
- Want keyboard interface
- Want POP3 and IMAP access

Sample - Remote email

- Solution for 50 users
  - RIM 950 $400 x 50 = 20,000
  - Access $50/mo x 50 = 2,500/mo
  - Training (seminars) = 4,000

Total First Year = $54,000

Sample - SFA

- Create an application tied to homegrown SFA software on a relational database.
- 50 users
- Simple user interface
- Multiple access - phone, Palm

Sample - SFA

- Thin Client Solution - PQA and WAP
- Login, simple queries with paging, note taking, record review
- 50 users

Sample - SFA

- Application development 150,000
  - New transactions, tools, environment hardware, software, people
- Devices (40 Palms w/ modems, 24,000
  10 Internet phones) 2,500
- Access (assuming voice today) 1,800/mo
- Training 10,000

Total First Year = $208,100

Sample - SFA

- C/S Solution - Native with Synch
- Login, simple queries with paging, note taking, record review
- 50 users

Sample - SFA

- Application development 350,000
  - New transactions, tools, environment hardware, software, people
- Devices (40 Palms w/ modems, 24,000
  10 Internet phones) 2,500
- Access (assuming voice today) 1,800/mo
- Training 10,000

Total First Year = $408,100

So, what's the good news?

- The costs are much less than two years ago when the equation looked like -
  - Application Development 500,000
  - Devices 200,000
  - Access 5,000/mo
  - Training 10,000

Total First Year = $770,000

Myth, Reality and History

- About interfaces
- About ‘simple’
- About standards
- About skills
- About technology today
- About price

About Interfaces

- Myth - Transcoding works. Some vendors will tell you that an XML definition will be able to be translated to multiple devices on different technology and devices (WML, HTML, HDML, PQA, cHTML).
- Reality - All devices have unique sizes and capabilities and transcoders are not substitutes for purposed design. It works for basic static content, not apps.
- History - If this was true, then we could write on screen and translate it to 3270, Windows, OS2 and HTML and have happy users. Yeah, right...

About ‘Simple’

- Myth - writing a wireless app is just like HTML/Web apps, except the screens are simpler and we use WML.
- Reality - Writing concise prose is harder than rambling pages. Program flow is more difficult.
- History - Writing tight applications on small footprint systems. An old grizzled veteran told me once, “The devil is in the details.” At 18 I did not realize how true that was. It is a rule that does not change regardless of the technology. Wireless delivery is ripe with details.

About Standards

- Myth - WAP is standard, so is Java.
- Reality - Nothing is standard.
- History - “Yeah, of course it will work on Netscape and IE.” “Yeah, if you write a GUI for OS/2 it’s a snap to go to Windows.”

About Skills

- Myth - Wireless skills are easy to find.
- Reality - Good wireless skills are damn near impossible to find.
- History - Any new technology has about a two to three year cycle before “real” engineers are in the market. The ones who have build real apps and not a ‘Hello World’ app.

About Price

- Myth - “I can do that fast, cheap and deliver quality.”
- Reality - Price, Quality and Speed. Pick any two...
- History - There is no free lunch in any business.

About Technology Today

- Myth - Current standards-based wireless technology can deliver robust applications.
- Reality - Stateless application are subject to many limitations (connectivity and speed).
- History - HTML and WAP apps have a major issue - disconnected operation (planes, subways, inside buildings). Native client/server apps with store and forward designs are better.

NotWired Products and Services

- Complete subscription front-office automation service for small and medium sized enterprises.
- Enterprise architecture and frameworks to deliver apps fast for IT shops.
- Enterprise services - wireless outsourcing, architecture, thin client, Palm, Java and distributed computing.

Business Products

- Complete groupware with wireless access.
- SFA with wireless access.
- Task management.
- Unified messaging.
- Access from browsers, Palm, PocketPCs, WAP phones and Native Palms w/ sync

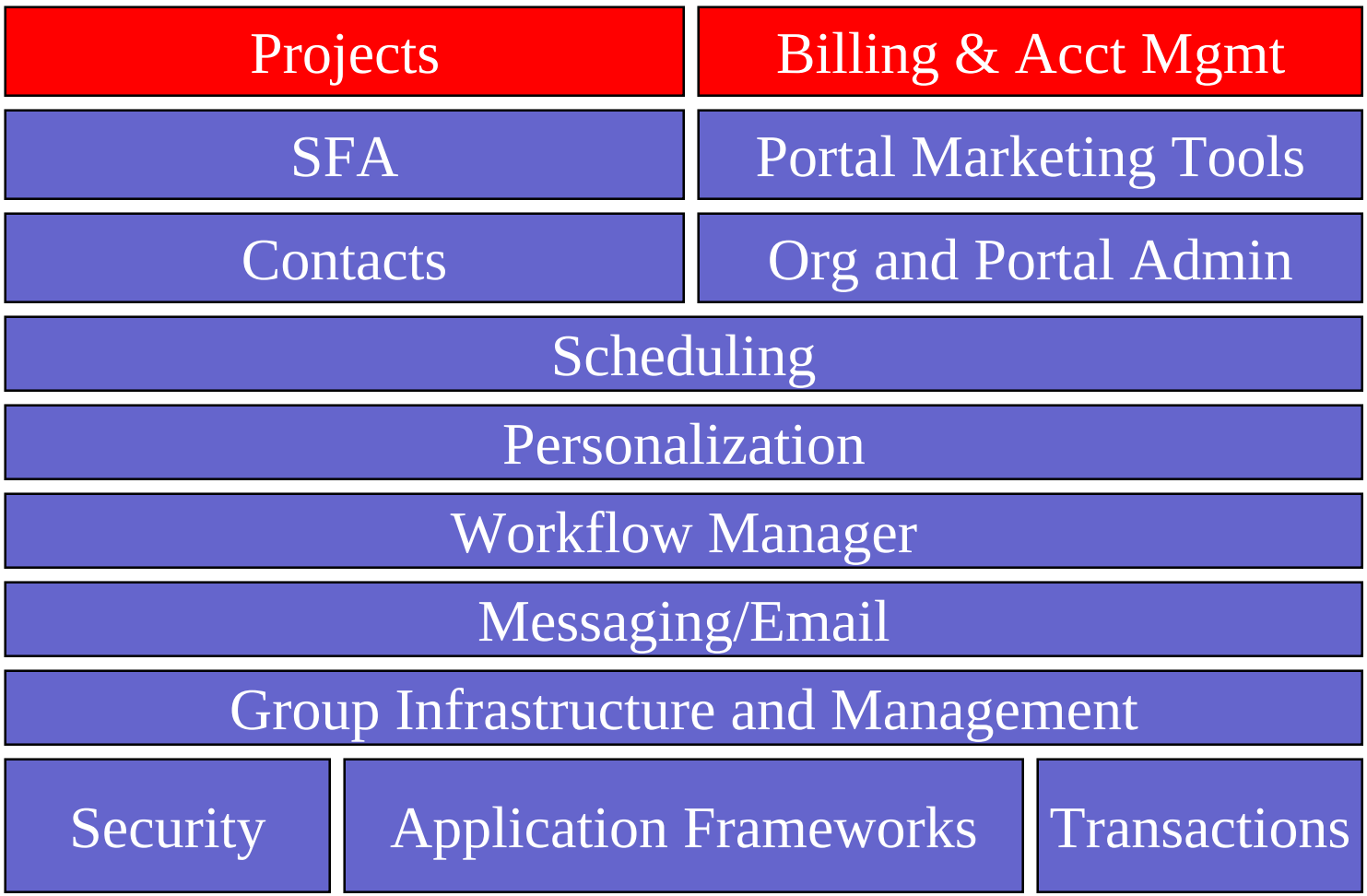
Enterprise Products

- Complete drop in architecture and frameworks.
- Complete development and build environment.
- Rapid, quality, reusable development.
- Skills to implement it fast.

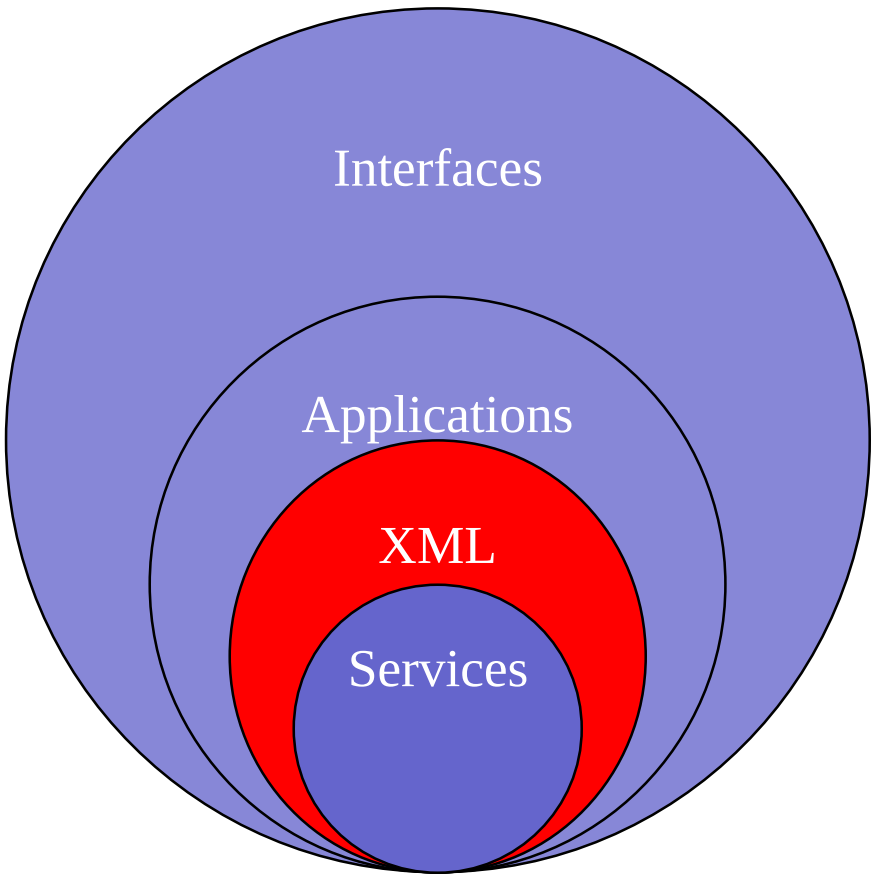
Enterprise Services

- Architecture Services
  - Development, Application and Deployment
- Design Services
  - Data, Object and UI
- Development
  - Infrastructure

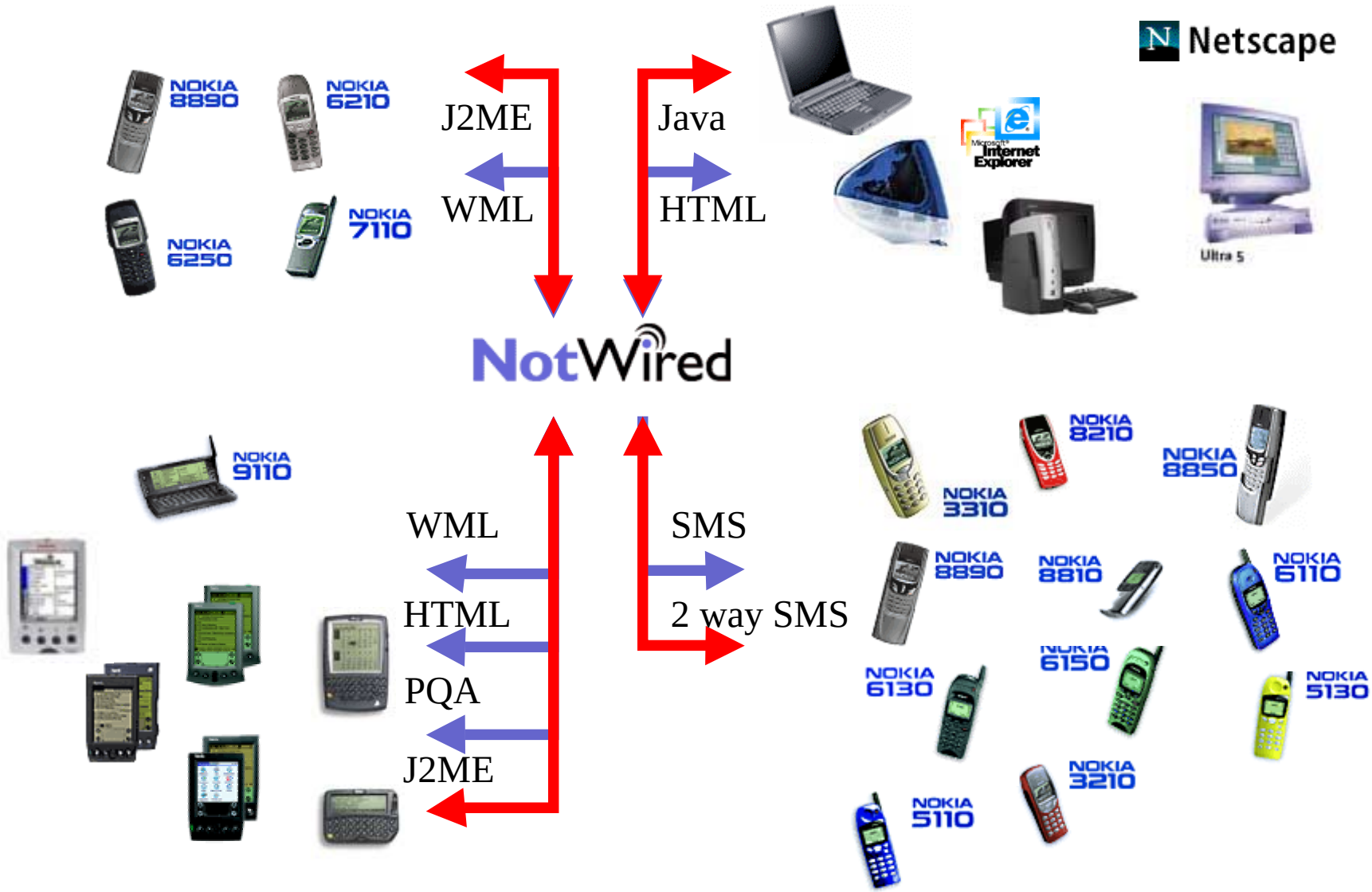
NotWired Components



Open

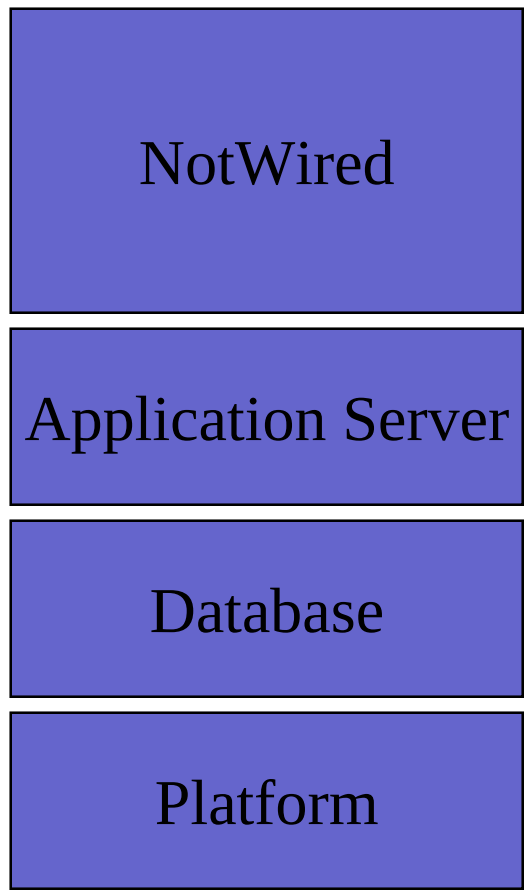


- Service layers
- Application layers
- Template based
- Standard language
- XML for service layer transactions


NOKIA 8890
NOKIA 6210
NOKIA 6250
NOKIA 7110
J2ME
WML
Java
HTML
Netscape
Microsoft Internet Explorer
Ultra 5
NotWired
NOKIA 9110
WML
HTML
PQA
J2ME
SMS
2 way SMS
NOKIA 3310
NOKIA 8210
NOKIA 8850
NOKIA 8890
NOKIA 8810
NOKIA 6110
NOKIA 6130
NOKIA 6150
NOKIA 5130
NOKIA 3210
NOKIA 5110

Standards

HTML
WML
XML
J2EE/JSP
J2ME
Apache
UNIX
PalmOS
PQA
SyncML





Apache, Websphere, WebLogic

Oracle, DB2, SQL Server





NotWired

“Empowering The Mobile Enterprise”



Contact Information

- Thank you!
- Kevin Clark
  - 312-266-7887 x105
  - kevin.clark@notwired.com
- Presentation at
  - http://www.notwired.com/presentations